

TurtleStitch

Making repeating designs and design for backing material

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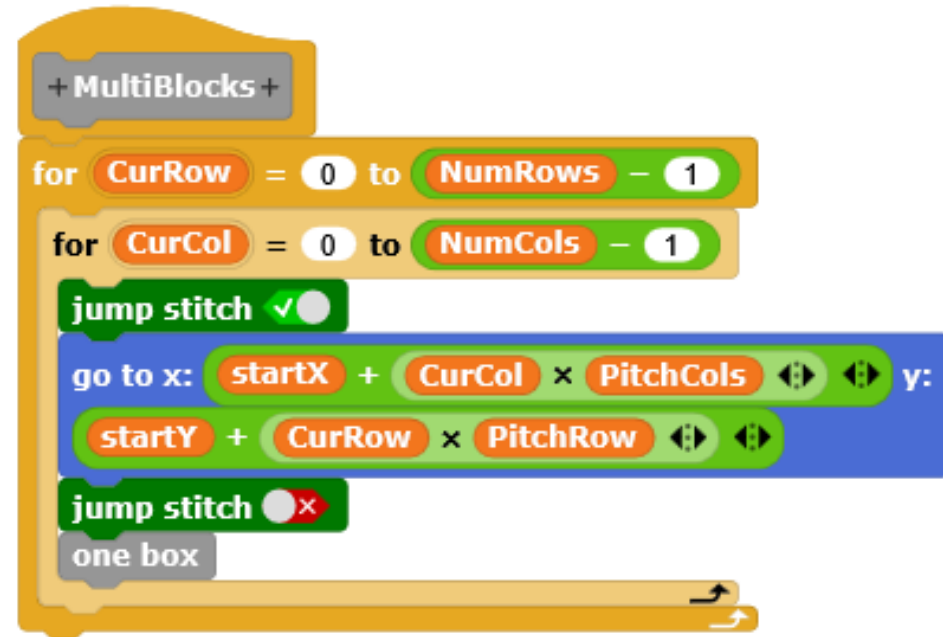
Making repeating designs



The original that
Susan Ettenheim requested
help for
(also note the jump stitches)

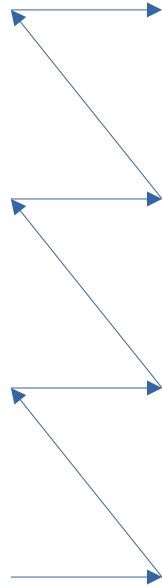
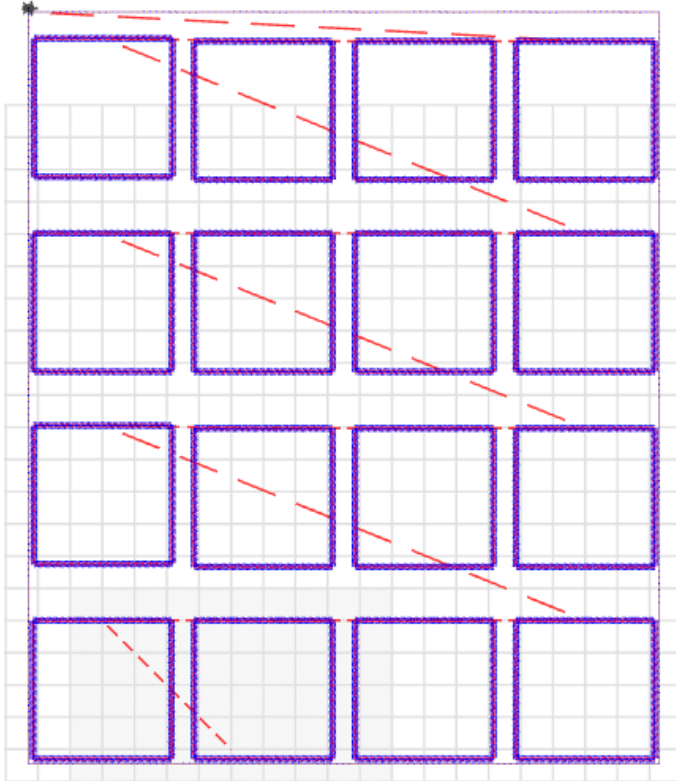
How to make designs with multiple repeating subdesigns

- 1) Make a single design that is only using relative coordinates (only relative to a startingpoint)
- 2) Use nested *for* blocks to create the multiple designs in rows and columns

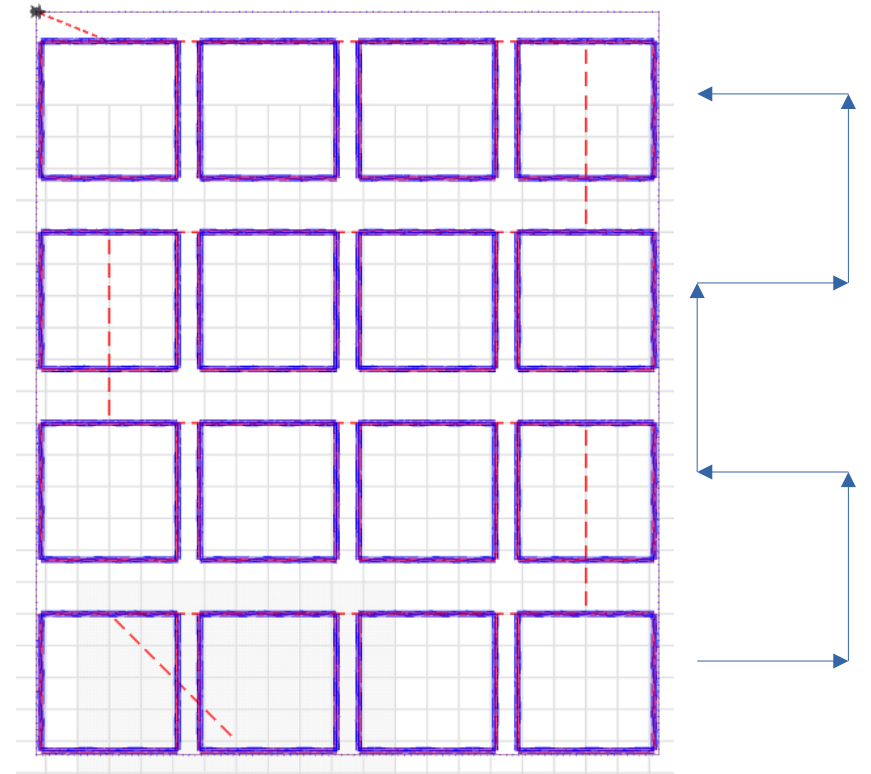


Avoiding long jump stitches

Instead of this ...



... do this



Therefore this
is the code for
moving through
rows and
columns



This if statement is added to limit the length of jump stitches by alternatingly go left to right and right to left for every other row

Using backing material: the intent

- Hide the jump stitches at the back of the fabric
- It goes like this:
 - Embroider the first part of the design
 - Let the embroidery machine stop
 - Tape a piece of fabric to the back of the hoop
 - Embroider a second part on both front and back together
 - Issues:
 - How to design the second part?
 - How to position it at the exactly the right place?

Get first and second design out of a single TurtleStitch program

- Use a switch in the design to determine which elements to include

```

+ one + box +
script variables TopX TopY
set TopX to x position
set TopY to y position
if OutlineOnly = False
  running stitch by 10 steps
else
  satin stitch with width 8 center
first_box
if OutlineOnly = False
  go to x: TopX + 0 y: TopY + 5
hole
pen up
go to x: TopX + -60 y: TopY + -50
pen down
hsfi
pen up
go to x: TopX + -50 y: TopY + -200
pen down
2030
pen up
go to x: TopX + 0 y: TopY + -80
pen down
heart

```

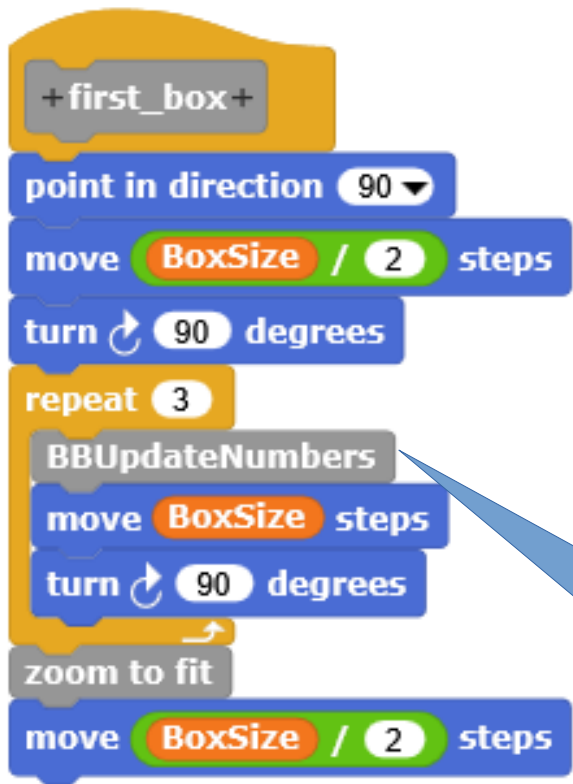
Remember the start position

Always embroider the box (*first_box*), the stitch type depends on OutlineOnly being True or False

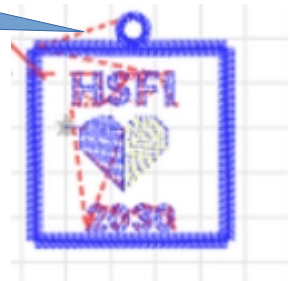
Skip the next elements if designing only the outline

Do all gotos relative to the start position

One more jump stitch avoidance: Start the box on the middle top

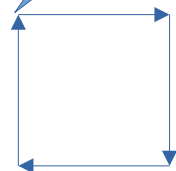


Avoid this
jump stitch

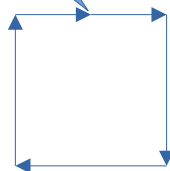


Start from the
middle top so that
the hole can be
done without a
jump stitch

Start and end



Original



New

We will talk about
BBUUpdateNumbers later

With the OutlineOnly flag the design can be steered

- It can be steered to create
 - The first part
 - The second part or
 - Both parts in a single design by changing *OutlineOnly* in the mean time.
 - Make the first design and the second design in separate colors
 - This makes the machine stop after the first design so that you can put the fabric to the back

Issue when making a separate first and second design

- The size of the first and second part will be different
 - Leading to the machine placing the design at slightly different places
 - Unless ...

Make a bounding box



Record the
minimum and
maximum
positions

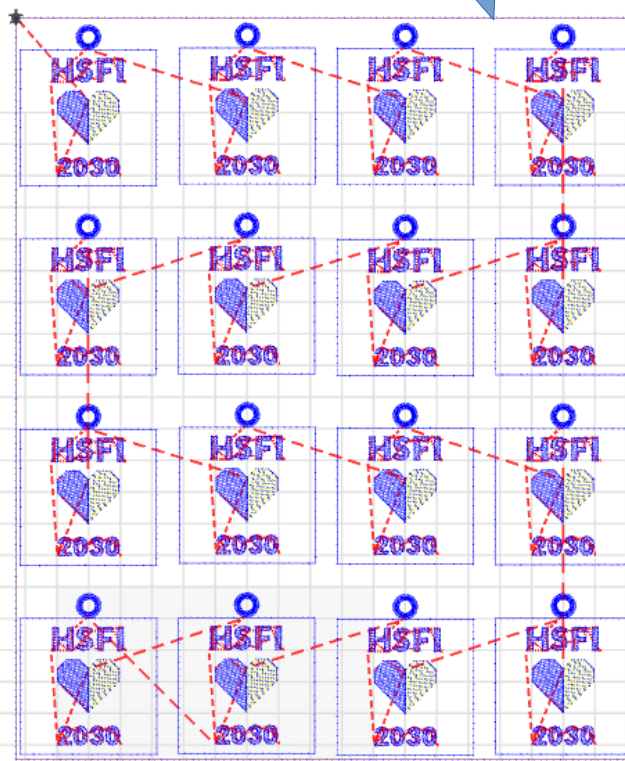
The bounding box at the outside is the same for both the first and the second design.

“Embroider” the bounding box in a different color so that the machine stops. Then do not embroider, but load the second design which will now be embroidered at the right place.

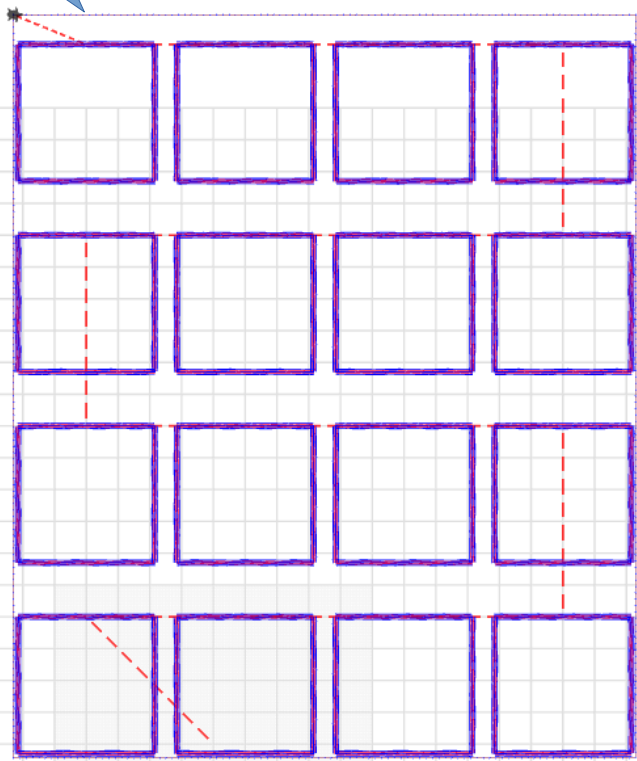
Same bounding
box

Result

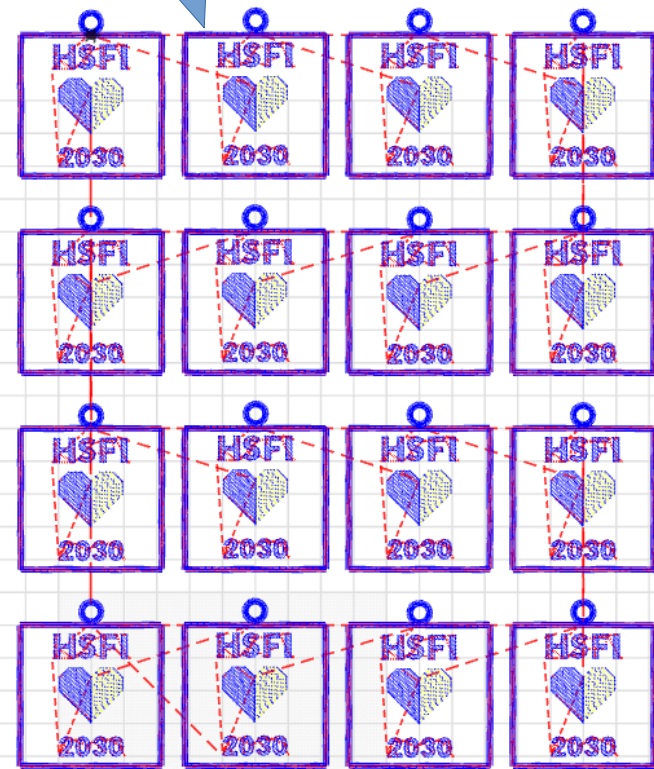
No bounding
box needed



First design



Second design



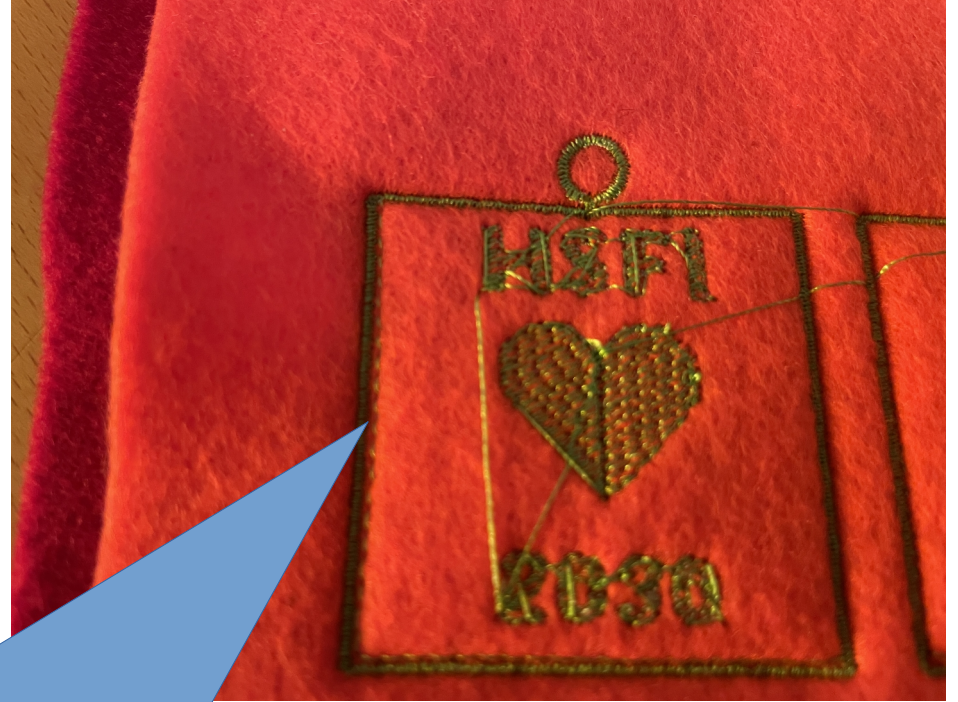
Combined design

Let's look at the embroidered results



Combined, result after the first part

Doing the back



This is a little off. Because putting felt at the bottom influences the top felt for a fraction of a mm

Alignment of first and second design



Perfect alignment
without doing anything
on the machine.

Only a few issues of a
fraction offset like in the
combined case due to
the fabric that was put
on the back.

Questions

- Is there a need for the box in the first design?
- The first design has multiple colors. Well prepared for a multi needle machine
 - On a single needle machine like mine I would have to change the color many times (e.g. 4 times per sub-design)
- This can be made better, let's however first draw conclusions from what was done so far

Conclusion

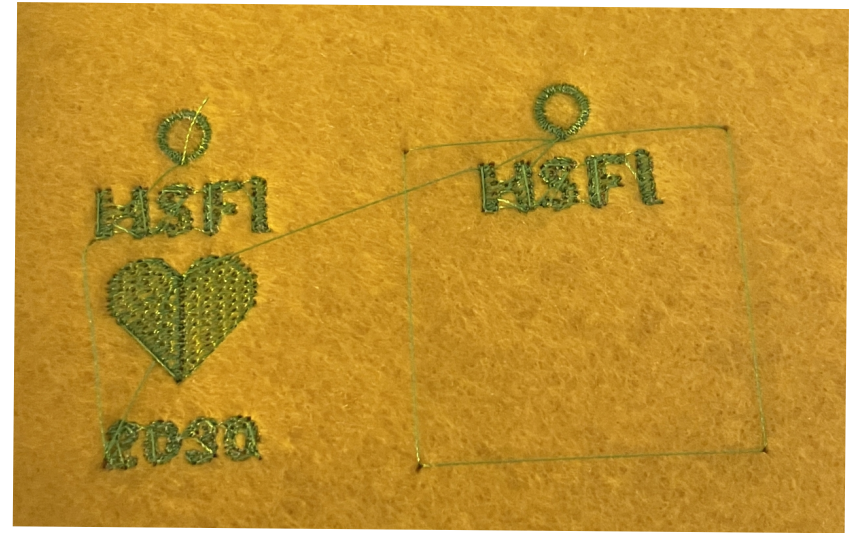
- For repeat designs, make one design with only relative coordinates
 - Then use nested *for* loops to create as many as you want
 - To reduce jump stitch length, consider how you jump from row to row
 - And look at the design to even further reduce jump stitches in the single design
 - The example works well for single color thread designs or for multi-needle machines
- When designing for backing material applied half way
 - Use the same program, but use a flag to include or exclude or change parts of the design
 - Use color change to stop the machine and then apply the backing and potentially load the second design
 - When splitting into two separate designs, make a bounding box around the designs that are exactly equal to let the embroidery machine position both designs in exactly the same position.
 - However, the result may be a fraction of a mm off due to the fabric on the back influencing the fabric on the front.

A failed attempt: Avoid the box in the first design

- Idea: do not sew the box in the first design, but make jump stitches instead to still record the positions for the bounding box

Result: not good

- Jump stitches do not only move the needle, but makes a stitch as well
- At the left box, the jump stitches are made, but the thread retracts.
- At the right box it makes actual stitches with long jumps



Next steps

- Avoid the box in the first part
 - Not needed as it will anyway disappear under the satin stitch of the second part
- Make it selectable that in the first part the patterns for a specific color of all sub-designs can all be embroidered together (color sorting)
- Find out why the ring is a few steps off in several of the sub-designs (must be a bug somewhere that I cannot find)
- Write documentation
- Anything else desired by anyone?

Questions?

Where to find the files?

- Original design:
https://turtlestitch.org/users/susanettenheim/projects/2030_HSFI
- Modified design:
https://turtlestitch.org/users/HansDeJongEhv/projects/2026_hsfi2026_2_boundinBox_WithOptions
- This presentation:
<https://github.com/hansdejongehv/TSUSB/blob/main/Miscellaneous/Repeated%20designs%20and%20backing.pdf>